Adani University

Bachelor of Technology

Subject Name: Data Communication & Computer Networks

Subject Code: EICCI24203 Sem: 4th Date: 02-02-2026

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EXPERIMENT: 2

OBJECTIVE:

To design and simulate a small computer network of 6-16-24 computers using switches and to verify successful data communication while ensuring availability of an alternative communication path. (With Ring Topology)

SOFTWARE USED:

CISCO PACKAGE TRAINER, DESKTOP

THEORY:

CODE:

Not applicable

PRACTICAL:

1. With 6 computers:

Figure 1A :- Basic Diagram

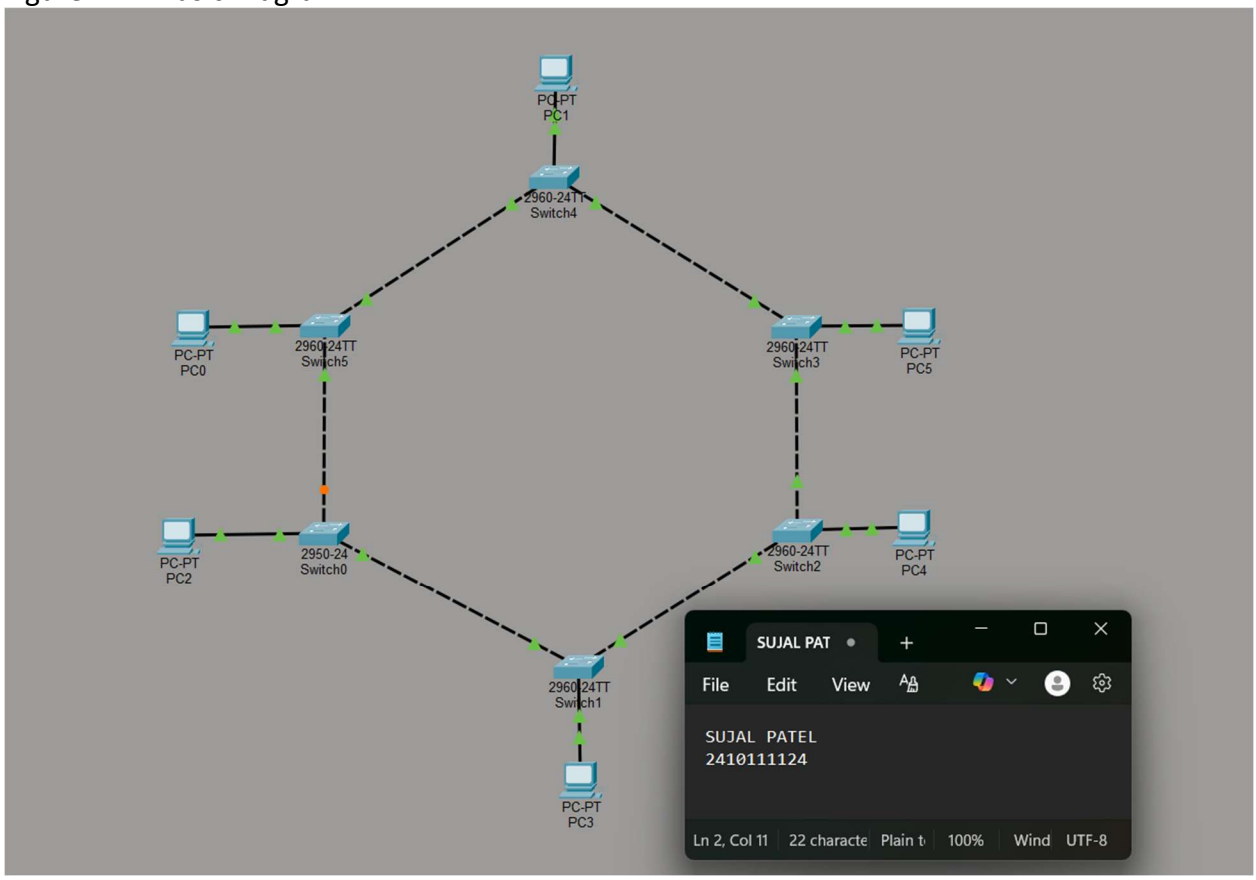


Figure 1B :- Simulation Diagram

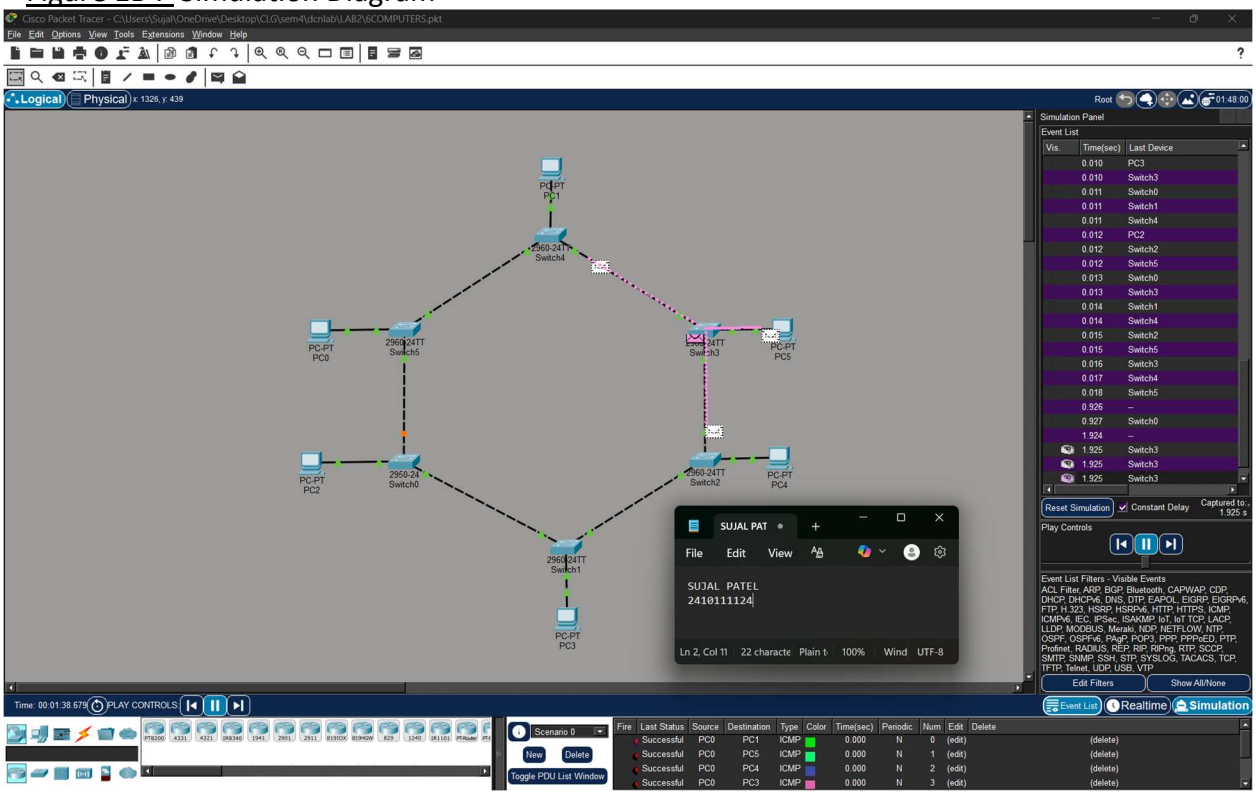
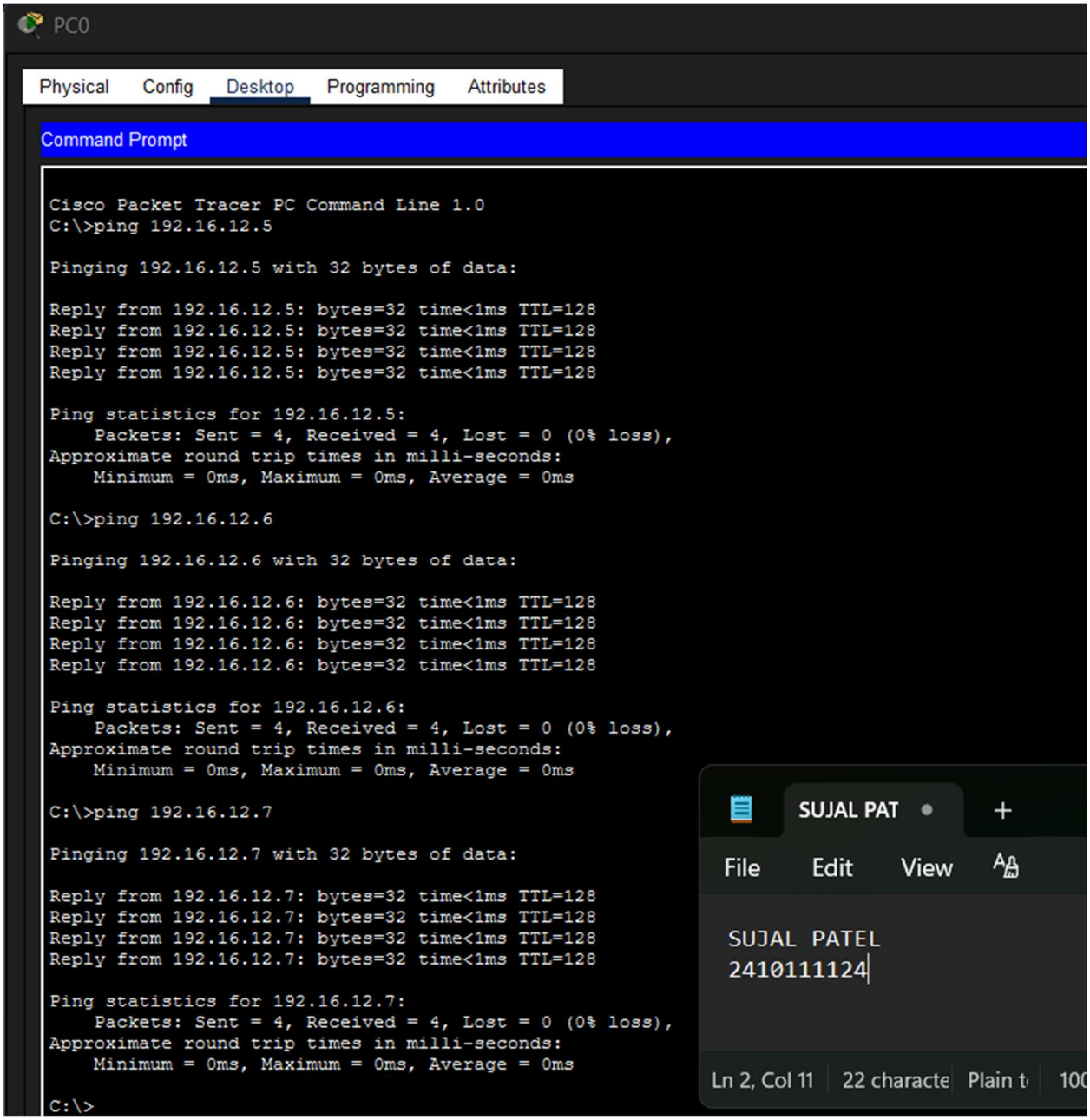


Figure 1C :- Analysis of message received



2.With 16 Computers :

Figure 2A :- Basic Diagram

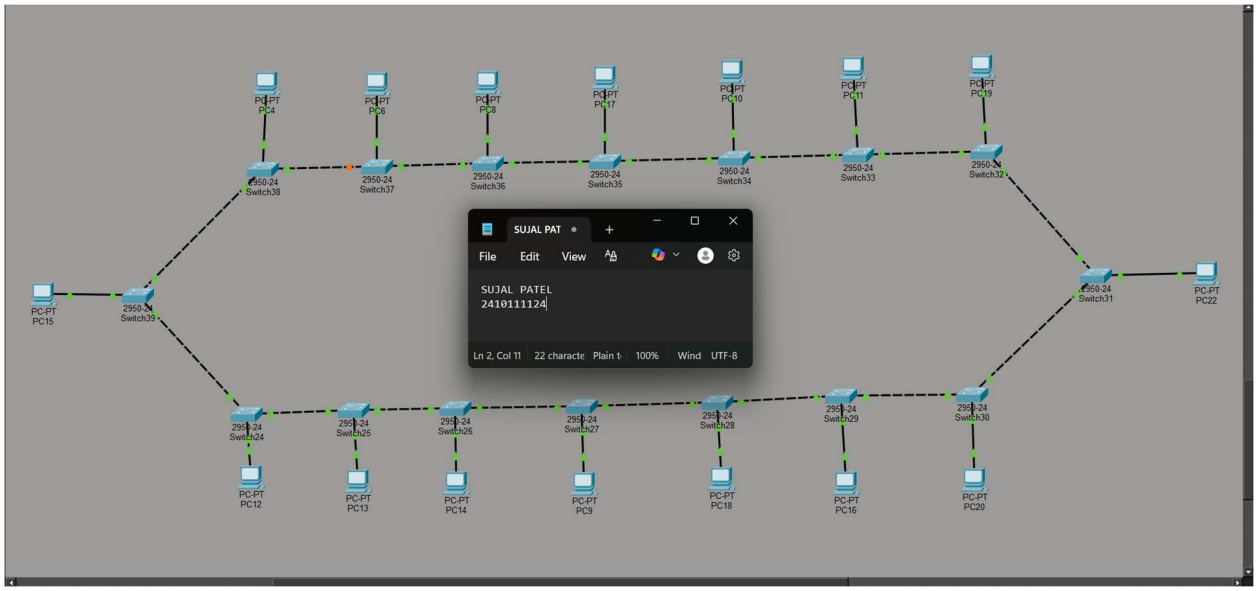


Figure 2B :- Simulation Diagram

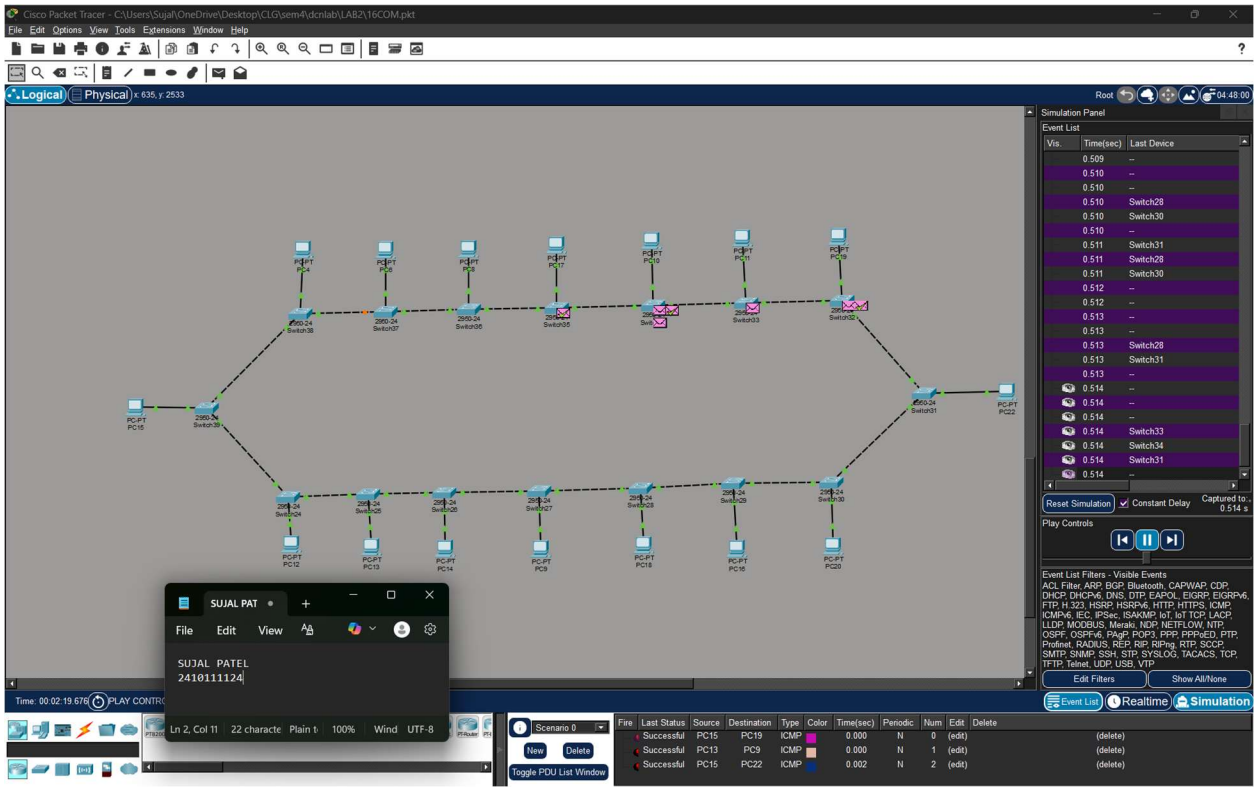
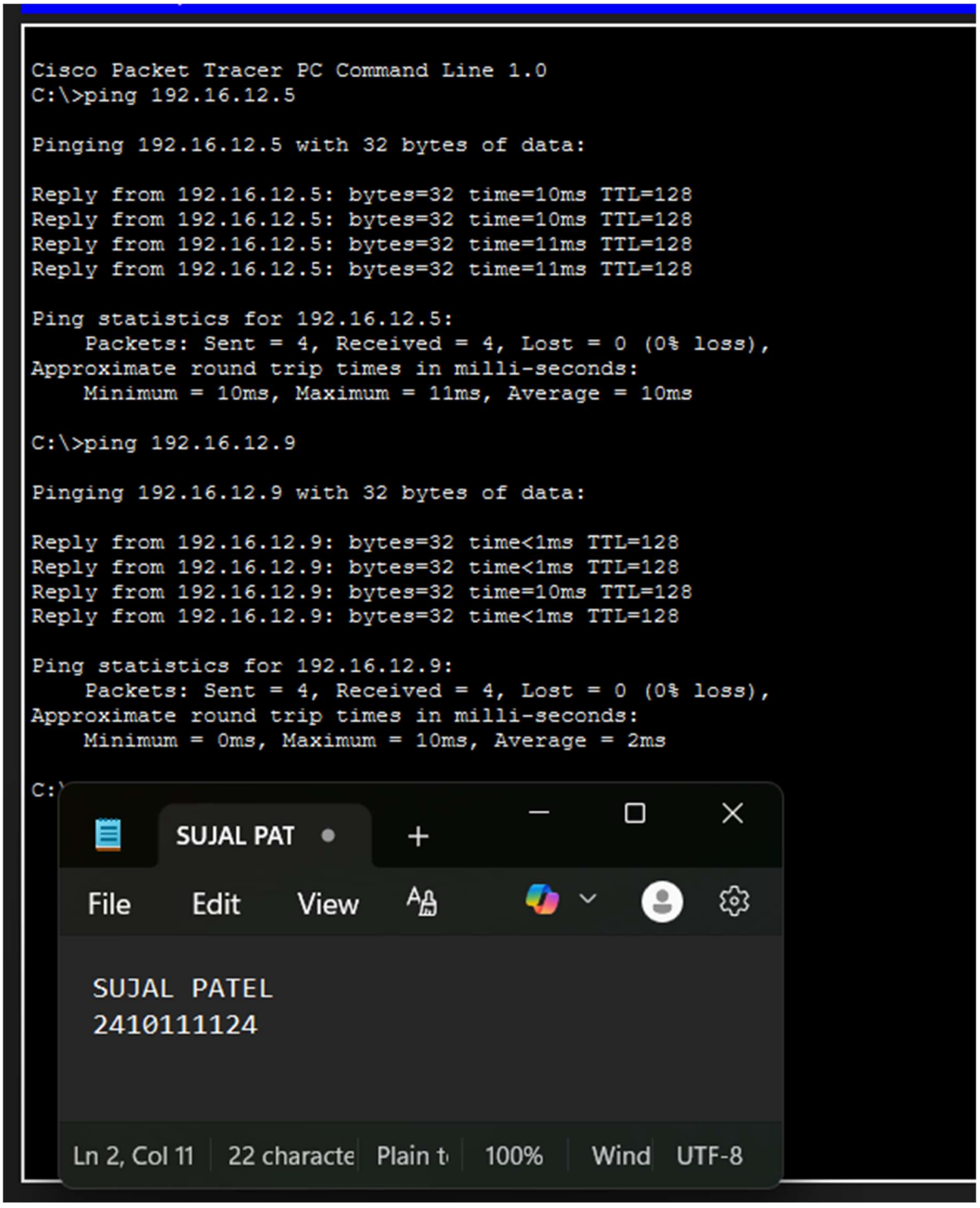
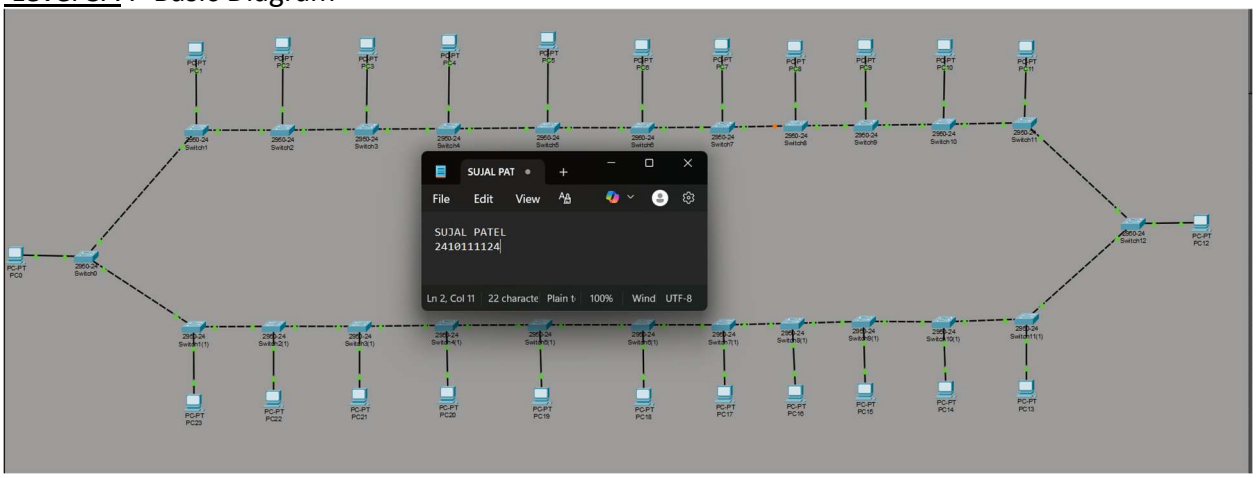


Figure 2C :- Analysis of message received

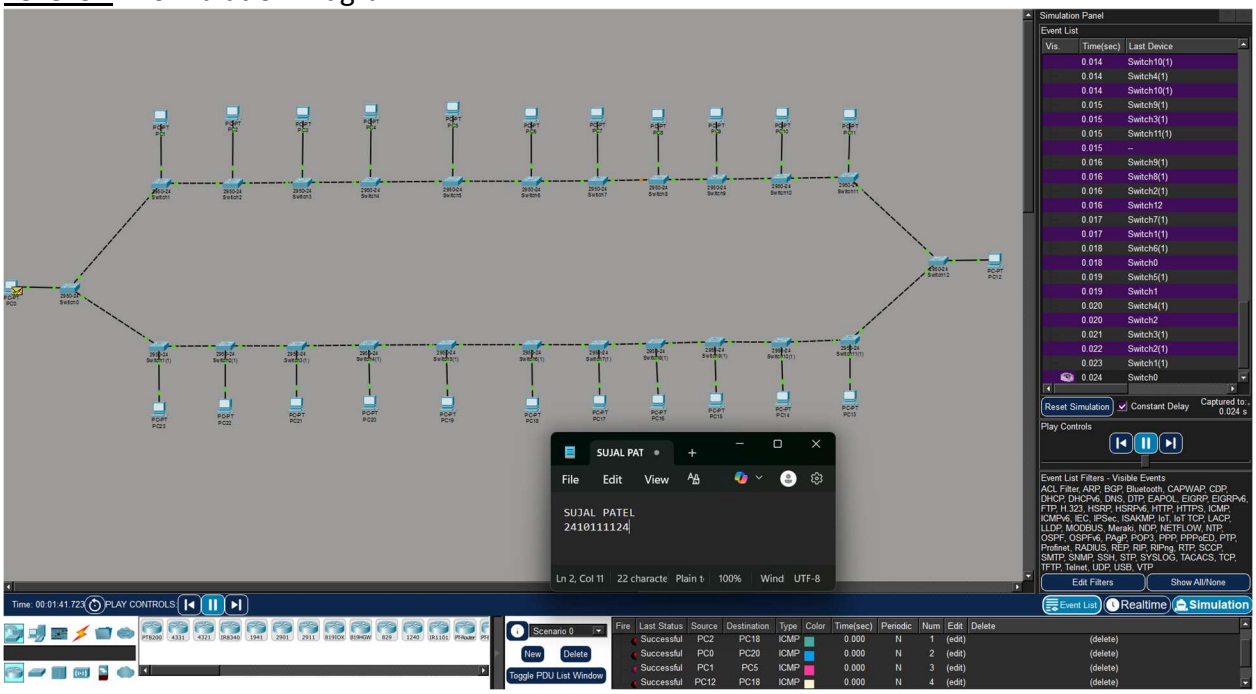


3.With 24 Computers

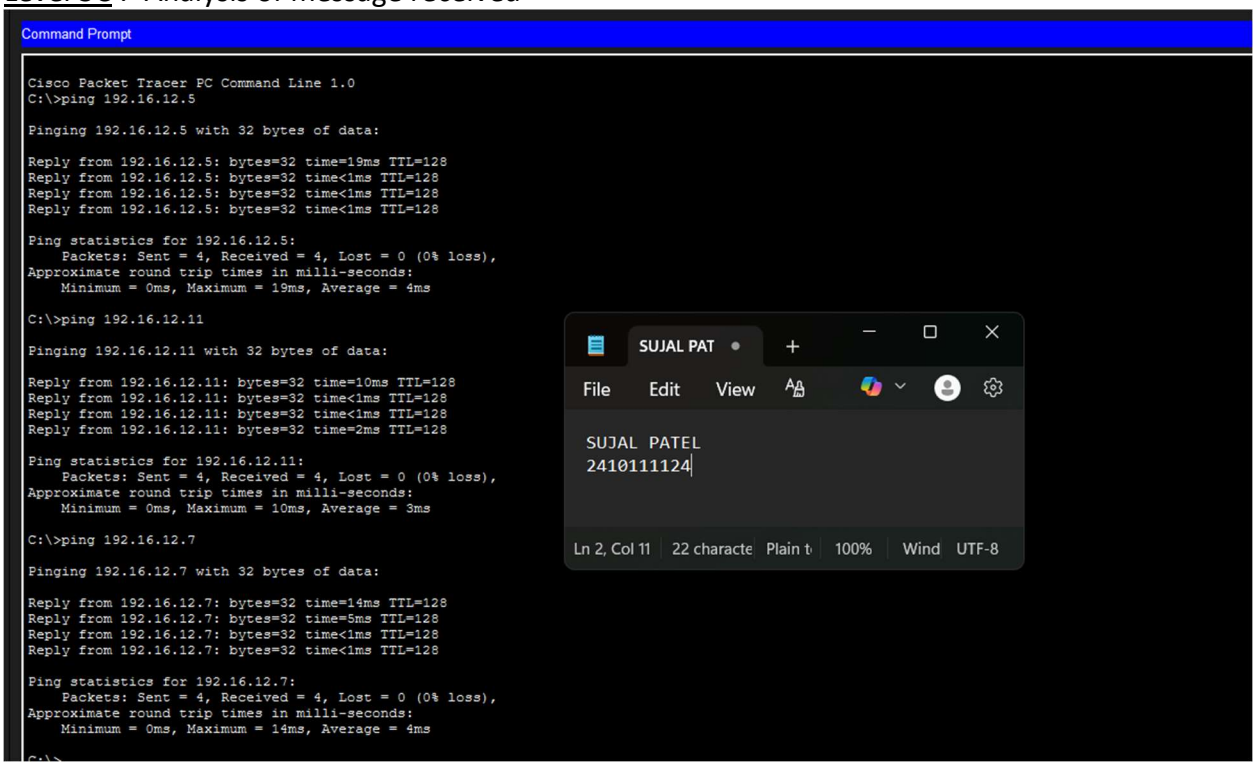
Level 3A :- Basic Diagram



Level 3B : - Simulation Diagram



Level 3C :- Analysis of message received



PROCEDURE:

1. Cisco Packet Tracer was launched and a new project was created.
2. The necessary number of PCs and switches were placed in the workspace.
3. Each PC was connected to a switch using cables.
4. The switches were linked together to create an alternative communication path.
5. Static IP addresses within the same network were assigned to all PCs.
6. The network was allowed to stabilize in real-time mode.
7. Message transmission was verified using simulation mode.

RESULTS:

The ring topology network consisting of 6 / 16 / 24 computers interconnected using switches was successfully designed and simulated.
All computers were able to communicate with each other without data loss.
When a single link or node failure was introduced, data transmission continued through the alternative path, proving the fault tolerance and reliability of the ring topology.

POSSIBLE SOURCES OF ERRORS:

1) Incorrect IP address or subnet mask configuration.
2) Entering IP address in the default gateway field instead of the IPv4 address field.
3) Improper cable connections between devices.
4) Testing the network before switch convergence was completed.

CONCLUSION:

The experiment successfully demonstrated the design and simulation of a ring topology network using switches.
The network ensured successful data communication among all nodes and maintained connectivity even after a link failure by providing an alternative communication path.
Thus, ring topology is suitable for networks where reliability and redundancy are required.

LEARNING OUTCOME:

1. How to design and simulate ring topology networks.
2. Configuration of IP addressing in a LAN.
3. Understanding of data transmission paths in ring topology.
4. Importance of alternative paths for fault tolerance.
5. How network topology affects performance and reliability.

PRECAUTIONS:

1) Ensure each PC has a unique IP address within the same subnet.
2) Use appropriate cable types for device connections.
3) Leave the default gateway empty when no router is used.
4) Allow sufficient time for the network to stabilize before testing communication